



VASUDHA SARAOGI

FOUNDER & CEO, NORTHSTAR
IMPACT SOLUTION

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**INDIA'S
TRANSITION FROM
REACTIVE CLIMATE
RESPONSES
TO PROACTIVE
MITIGATION
HINGES ON HOW
INTELLIGENTLY
WE INTEGRATE
TECHNOLOGY
ACROSS SECTORS**

TECH FOR SUSTAINABILITY

From real-time emissions monitoring to AI-driven resource analytics, technology is shifting the sustainability agenda

SUSTAINABILITY IN INDIA can no longer be treated as a cyclical discussion tied to annual reports and regulatory deadlines. Sustainability must become a continuous, intelligence-driven discipline.

For India, where economic advancement intersects with urgent decarbonisation imperatives, this evolution is increasingly enabled by one transformative force: technology.

India is emerging as a crucible of tech-enabled sustainability. From real-time emissions monitoring to AI-driven resource analytics, technology is shifting the sustainability agenda.

Technology is enabling a shift. Remote sensing, IoT sensors, and AI analytics now power real-time Monitoring, Reporting, and Verification (MRV) systems, flagging emissions, water, and air quality anomalies for timely response. Yet fewer than 10% of industrial units operate continuous monitoring. The imperative is clear—scaling these systems isn't just a technical upgrade, it's a national resilience strategy.

FROM OBSERVATION TO PREDICTION

One of the most valuable contributions of technology to sustainability is its capacity to unify granular operational data with planetary-scale environmental intelligence.

At the micro level, digital infrastructure enables organisations and farmers to optimise electricity, water, and materials in real time. The results are evident.

At the macro level, satellite imagery, earth observation systems, and advanced climate models offer a panoramic view of ecological change, whether deforestation, glacial retreat, or urban heat stress.

India's transition from reactive climate responses to proactive mitigation hinges on how intelligently we integrate technology across sectors.

RISE OF CLIMATE TECH

India's climate technology growth is supported by structural enablers and a receptive policy environment. Additionally, India is implementing a digital-first carbon market framework designed to include SMEs, which promotes accountability through technology-enabled MRV. With projections of over two million green jobs by 2030 across the circular economy, India's climate tech surge is not just ecological but economic.

Digital platforms are also unlocking cross-sector collaboration, benchmarking, and decentralised innovation—reducing silos. Despite its transformative potential, integrating technology into sustainability efforts is not frictionless. On-the-ground deployment reveals